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Summary

Engineer working on clinical genomics infrastructure: workflow orchestration, deployment automation, and reproducibility for scientific computing. Production experience running a regulated clinical variant review platform across five environments, plus open-source contributions on local-first pipeline orchestration across five workflow engines.

Experience

Software Development Engineer, NonStop io Technologies

May 2026 – Present

Clinical Genomics Platform Engineering | Pune, India (Hybrid)

- Owned maintenance and deployment of **backend**, a FastAPI/PostgreSQL clinical variant review platform, across **5 environments** (dev, QA, UAT, staging and prod) on Oracle Linux 8.10 with systemd, nginx, uvicorn, SELinux, and Azure Repos/Pipelines.
- Authored **genie.sh**, a 14-phase resumable bootstrap and deployment script with SELinux/firewalld handling and GIT_ASKPASS credential management; encodes hard assertions derived from **5 prior production incidents**, reducing environment setup from 3 hours of manual steps to a single command.
- Diagnosed and resolved **5 production incidents** on QA/UAT, including an nginx named-location bug that discarded URI paths and returned **404 across the entire API**, an Entra v1/v2 token mismatch causing HTTP 401s, and a cases_users primary-key mismatch requiring a live ALTER TABLE on a clinical database.
- Built a Swagger-driven validation harness covering all **65 endpoints** of the APIs, catching contract regressions before UAT sign-off.
- Designed CI/CD architecture spanning the same five environments.
- Delivered handover documentation (**DEPLOYMENTS.md**) and coordinated **OpenTelemetry** instrumentation rollout with the observability team.
- Built **InterGenix**, a healthcare integration service (**Spring Boot 3, Java 17, HAPI HL7v2**) acting as the validation and orchestration layer alongside Mirth Connect, with Medplum FHIR integration and OpenELIS polling; pipeline reached full operational status.

Software Development Engineer Intern, NonStop io Technologies

Feb 2025 – May 2026

Data Engineering, ML Pipelines, Backend Systems

- Designed and executed genomics data workflows handling **100–300 GB datasets**, with emphasis on structured processing and pipeline reliability.
- Built workflow orchestration, run tracking, and event-based monitoring services using REST APIs and **PostgreSQL JSONB** for semi-structured run state.
- Automated multi-step execution pipelines with failure recovery and re-run logic, removing manual intervention from repetitive workflow reruns.
- Deployed and executed workflows across local environments and **AWS Batch, EKS, and HealthOmics**.
- Built an **LLM-assisted analysis system** generating actionable summaries from structured reports, reducing manual analysis turnaround.
- Developed microservices in **Spring Boot and FastAPI**; worked with Python, SQL, Docker, JMeter, RabbitMQ, and Git in an agile environment.

Projects

StrixFlow : Local-first genomics workflow orchestration platform

Tech Stack: Python, Nextflow, Snakemake, CWL, WDL, Playwright, ruff

- Orchestrates **5 workflow engines** (Nextflow, Snakemake, CWL, WDL, custom Python) without pipeline rewrites, targeting clinical settings where genomic data cannot leave the institution.
- Recovers DAG structure via **native engine introspection** rather than reimplemented parsers, so the displayed graph matches the graph the engine executes.
- Validated by **395 passing unit tests** plus Playwright end-to-end coverage of execution and path-based pipeline registration flows.

GraphLM : Knowledge-graph-augmented retrieval for document understanding

Tech Stack: FastAPI, React, PostgreSQL, Neo4j, Qdrant

- Combined **RAG, vector search, and knowledge graphs** to support multi-hop contextual retrieval over unstructured research documents.
- Basis for a published survey on knowledge graphs for intelligent document understanding.

ADAPT : Alzheimer's Disease Analysis and Prediction Tool

Tech Stack: Python, VGG-19, DVC, MLflow, Docker, FastAPI, React

- Built an end-to-end MRI classification pipeline using **VGG-19**, with **DVC** for data and model versioning and **MLflow** for experiment tracking.
- Containerized with Docker for reproducibility; deployed FastAPI backend on Google App Engine and React frontend on Netlify.

MetaboliQ AI : ML platform for diabetes risk prediction

Tech Stack: Python, Scikit-learn, Flask

- Built predictive models reaching ~**83% accuracy** across a full preprocessing, training, and evaluation workflow.
- Led a team of 4 as project lead; selected **Top 10 Finalist, Intel AI Hackathon (IEEE INDICON, IIT Kharagpur)**.

Publications

- **GraphLM: A Comprehensive Survey on Knowledge Graphs for Intelligent Document Understanding.**
<https://doi.org/10.22214/ijraset.2025.75390>
- **Advancing Neurosymbolic AI: A Comprehensive Review of Hybrid Reasoning Frameworks and Applications.**
<https://doi.org/10.55248/gengpi.6.0525.1613>

Technical Skills

Infrastructure and Deployment: Linux (Oracle Linux, systemd, SELinux, firewallld) | nginx | Docker | Kubernetes | AWS (Batch, EKS, HealthOmics) | Azure Pipelines | Bash | CI/CD

Workflow Orchestration and MLOps: Nextflow | Snakemake | CWL | WDL | Apache Airflow | DVC | MLflow | OpenTelemetry

Bioinformatics and Health Data: Clinical genomics pipelines | HL7v2 (HAPI) | FHIR (Medplum) | Mirth Connect | OpenELIS

Languages and Backend: Python | Java | SQL | JavaScript | FastAPI | Spring Boot | REST APIs | System Design

Data and ML: PostgreSQL (incl. JSONB) | Databricks | Snowflake | dbt | Neo4j | Qdrant | Scikit-learn | PyTorch | TensorFlow | Hugging Face Transformers

Education and Languages

B. Tech, Artificial Intelligence and Data Science,
AISSMS Institute of Information Technology, Pune

- **CGPA: 8.36/10.** Coursework: Machine Learning, Deep Learning, NLP, MLOps, Operating Systems, DBMS, Cloud Computing, Data Structures.

Achievements

- HSC Rank 1
- Top 10 Grand Finalist, Intel AI Hackathon (IEEE INDICON, IIT Kharagpur)
- First Runner-Up, ISB&M Pune Analytics Case Study Competition
- Most Innovative Award, QA AI Agent Hackathon 2026

Languages : English, Hindi, French